

# Dense point spectrum forces a nontrivial closed invariant subspace

Conjecture 00000000928 - FALSE | SucRunBug | 2 October 2026

## Filed assertion

A Hilbert-space operator has point spectrum dense in the unit circle, full spectrum equal to the disk, and no invariant subspace. Use the standard convention: a bounded complex linear operator and a nonzero proper closed invariant linear subspace.

## Closed eigenspaces

Let  $P(T)$  be the set of scalars  $\lambda$  admitting a nonzero vector  $v$  with  $Tv = \lambda v$ . If  $T$  has no nonzero proper closed invariant subspace, every eigenvalue has eigenspace  $E_\lambda = \ker(T - \lambda I)$  equal to the whole space. Indeed, this kernel is closed by continuity, is nonzero by the eigenvalue definition, and is invariant since  $T(Tv) = \lambda Tv$ .

## At most one eigenvalue

For eigenvalues  $\lambda$  and  $\mu$  choose a nonzero  $v$  in  $E_\lambda$ . Because  $E_\mu$  is also the whole space,  $\lambda v = Tv = \mu v$ , giving  $\lambda = \mu$ . Thus  $P(T)$  has at most one element. A set with at most one element in the complex plane is closed, and its closure still has at most one element.

## Density contradiction

Density on the unit circle forces the closure to contain both 1 and -1, two distinct points. This is impossible. Therefore density alone forces a nonzero proper closed invariant subspace. Adding the requirement that the full spectrum equal the disk cannot eliminate the contradiction.

## Formalization and scope

Lean uses the actual Mathlib eigenvalue predicate and actual closed eigenspaces of continuous linear maps. The density hypothesis only asks that the unit circle be contained in the closure of the point spectrum. The main theorem produces a submodule that is closed, nonzero, proper, and invariant. The proof works on all complex normed spaces and makes no claim to resolve the general invariant subspace problem.

## Reproduction

In lean4/ run lake exe cache get, lake build, then lake env lean Check.lean. Lean/Mathlib v4.33.0; pinned Mathlib commit db584cd6d46c92f209a44c0f1c829460d327499d. Actual check outputs and source hashes are recorded in verification.txt.

Source: official conjecture 00000000928.md at commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.